

0 2 . 1

Put a tick (✓) in the box next to the equation that shows how the net production of consumers, N , can be calculated where

I represents the chemical energy store in ingested food

F represents the chemical energy lost to the environment in faeces and urine

R represents the respiratory losses to the environment.

[1 mark]

$$N = (I - F) + R$$

☐

$$N = I - (F + R)$$

☐

$$N = I + (F + R)$$

☐

$$N = I - (F - R)$$

☐

In the UK, some female cattle are only used for breeding. This female breeding herd has dairy cows and beef cows.

Table 1 shows data on dairy cows and beef cows in the UK female breeding herd in December 2013 and December 2017.

Table 1

Date	Total number in female breeding herd / millions	Percentage of total female breeding herd	
		Dairy cows	Beef cows
December 2013	3.35	54	46
December 2017	3.45	55	45

0 2 . 2

In December 2017, the female breeding herd was 48% of all female cattle in the UK. Use **Table 1** to calculate the percentage of all female cattle that were beef cows in the UK in December 2017.

[1 mark]

Answer _____ %

0 2 . 3

Use **Table 1** to calculate the increase in the number of dairy cows in the UK female breeding herd between December 2013 and December 2017.

Show your working.

[2 marks]

Increase in number _____

0 2 . 4

Farming cattle for humans to eat is less efficient than farming crops because of energy transfer. Explain why.

[2 marks]
